In [15]:
```
import pandas as pd

df = pd.read_excel('DataMajorCompagnies.xlsx')
df = df.sort_values(['Company', 'Year']).reset_index(drop=True)

df['Revenue Growth (%)'] = df.groupby('Company')['Total Revenue'].pct_change() *
df['Net Income Growth (%)'] = df.groupby('Company')['Net Income'].pct_change() *
df['Assets Growth (%)'] = df.groupby('Company')['Total Assets'].pct_change() * 1
df['Liabilities growth (%)'] = df.groupby('Company')['Total Liabilities'].pct_ch
df['Cash flow growth (%)'] = df.groupby('Company')['Cash Flow'].pct_change() * 1

df.round(2)
```

Out[15]:

	Company	Year	Total Revenue	Net Income	Total Assets	Total Liabilities	Cash Flow	Revenue Growth (%)	Net Income Growth (%)	As Gro
0	Apple Inc.	2023	383.28	97.00	352.58	290.44	110.54	NaN	NaN	
1	Apple Inc.	2024	391.04	93.74	364.98	308.03	118.25	2.02	-3.36	
2	Apple Inc.	2025	416.16	112.01	359.24	285.51	111.48	6.43	19.50	
3	Microsoft	2023	211.92	72.36	411.98	205.75	87.58	NaN	NaN	
4	Microsoft	2024	245.12	88.14	512.16	243.69	118.55	15.67	21.80	2
5	Microsoft	2025	281.72	101.83	619.00	275.52	136.16	14.93	15.54	2
6	Tesla	2023	96.77	15.00	106.62	43.01	13.26	NaN	NaN	
7	Tesla	2024	97.69	7.09	122.07	48.39	14.92	0.95	-52.72	1
8	Tesla	2025	94.83	3.79	137.81	54.94	14.75	-2.93	-46.50	1

To analyse and compare the financial performance of Apple, Microsoft, and Tesla from 2023 to 2025: I imported dataset from excel into Python using pandas then I calculated annual growth percentages for key indicators.

1- Microsoft: Shows steady, reliable growth across the board. Both revenue and net income increase consistently year-over-year (Roughly 15% and 14% growth). It is a strong business stability. 2- Apple: Experienced a slight slowdown in 2024, but bounced back in 2025. Net income surged by nearly 20% and the company optimized its balance sheet by reducing liabilities and total assets. 3- Tesla: Asset growth continues but profitability is facing heavy pressure.Revenue dipped slightly in 2025 and net incope dropped significantly in both 2024 (-52.72 %) and 2025 (-46.5%).

Conclusion: Microsoft stands out as the most balanced and consistently performant company over this period. Apple demonstrates a strong capacity for recovery whereas Tesla struggles with maintaining profit margins despite ongoing asset expansion.

In [5]:
```
%pip install openpyxl
```

```
Requirement already satisfied: openpyxl in c:\users\elias\appdata\local\programs
\python\python312-arm64\lib\site-packages (3.1.5)
Requirement already satisfied: et-xmlfile in c:\users\elias\appdata\local\program
s\python\python312-arm64\lib\site-packages (from openpyxl) (2.0.0)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 25.0.1 -> 26.2.2
[notice] To update, run: python.exe -m pip install --upgrade pip
```